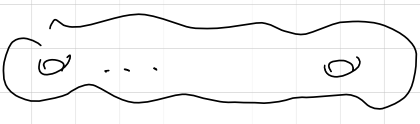
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



(n≥1)
↓
$\pi_1=0$.

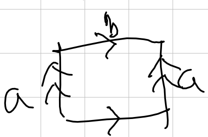
2-dim mfd

$L(p,q)$

$M \# N$. ∧ ∨

$S^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

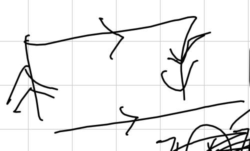
$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.

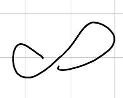


Klein. free product.

abababb..

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

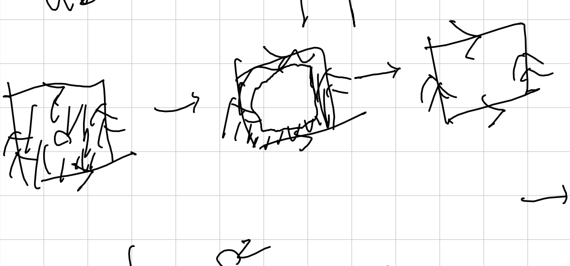


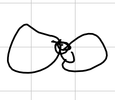
$\langle a,b \mid ab=ba\rangle$

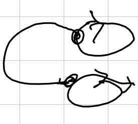
$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

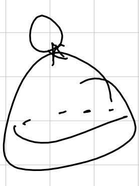
$ab=ba \to$ grup



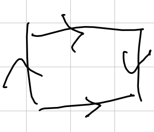


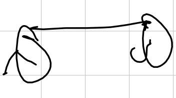


$S^1\vee S^2$



T: $\mathbb{Z}\times\mathbb{Z}$





$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

$T\setminus\bullet$: $\mathbb{Z}*\mathbb{Z}$

$K\setminus\bullet$: $\mathbb{Z}*\mathbb{Z}$

theorem

S^3: 0.
↓
$\mathbb{R}^3$: 0.

Theorem

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

S^1
↕
$\mathbb{Z}$

M: n-dim
with (n≥3)

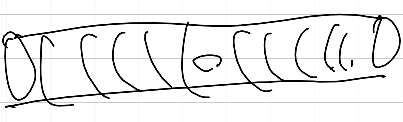
then $\pi_1(M)\simeq\pi_1(M\setminus\bullet)$.

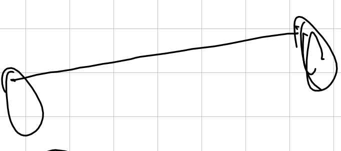
$\simeq\pi_1(M\setminus$ 2pt.

$\simeq\pi_1(M\setminus\{x_n\})$.

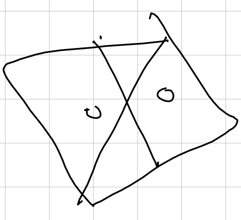
$\pi_1(S^n)=0$
n≥2.

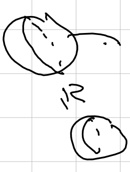
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$
(n≥2).



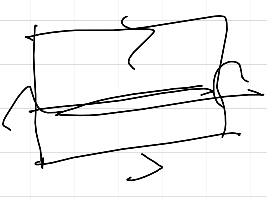


tree



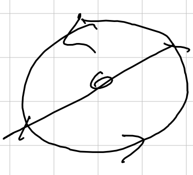


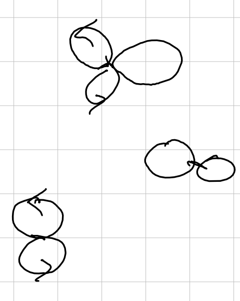
$S^1\vee S^1\vee\mathbb{R}$.
↓
$\mathbb{Z}*\mathbb{Z}$



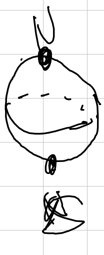
: $S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

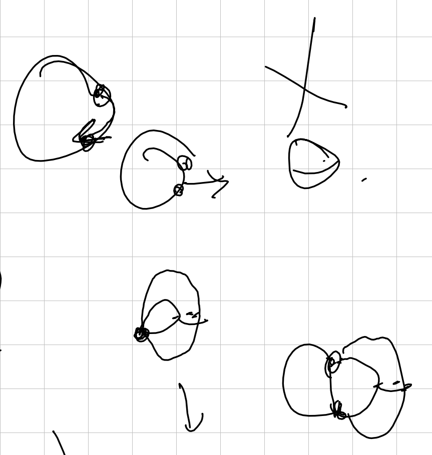


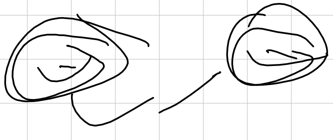


Example.

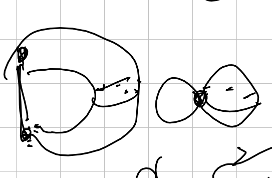


$S^2/N=S$

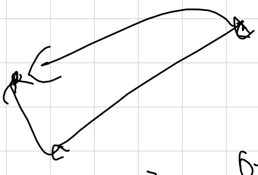


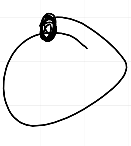


$\to\pi_1(\ldots)$



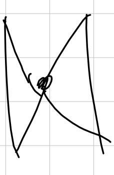
$S^1\vee S^2$.





S^1: 0-cell 1개
1-cell 1개.

S^2: 0-cell 1개
2-cell 1개



$S^1\vee S^1$

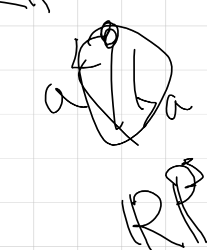
$\mathbb{RP}^2$: 0-cell: 1개
1-cell: 1개
2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\langle\ldots\rangle$

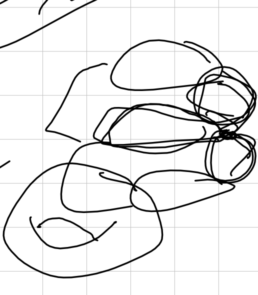
$\mathbb{RP}^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$
$a^2=1$

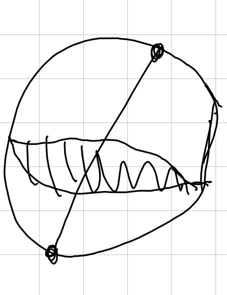


$\mathbb{RP}^3$: ~~0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

~~$\pi_1(\ldots)$~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

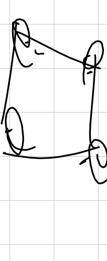


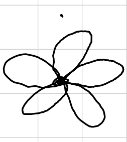


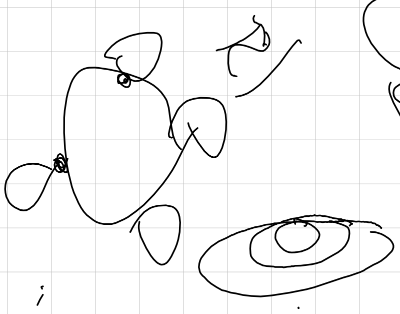
←

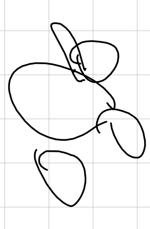
$x^2+y^2+z^2=1$
∪ $x^2+y^2\le1$
∪ $(0,0,z)$ $-1\le z\le1$.

$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$

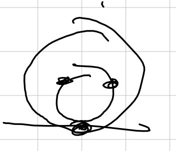


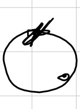


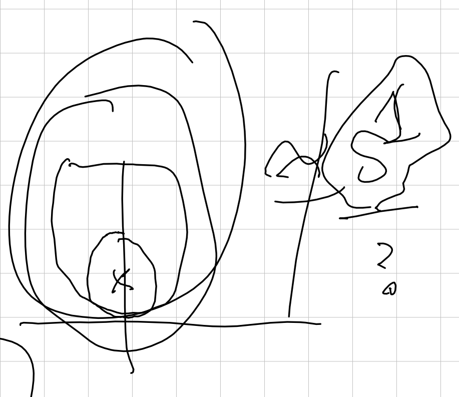


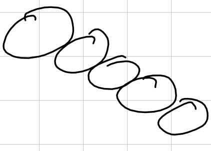


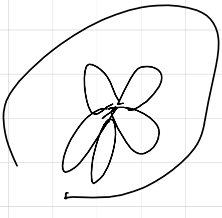
$\mathbb{R}/\mathbb{Z}$

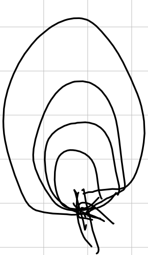


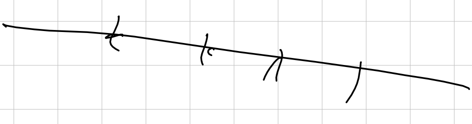


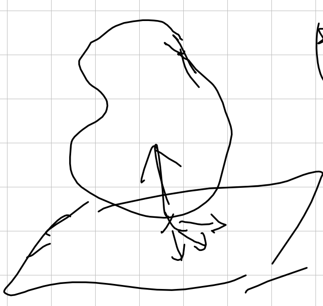




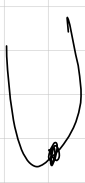








$r=0$



$z=x^4+y^4$
↓ exp

$K_1=f(y,z)$..
$K_2=f_y$..

$\frac{1}{p_n+z}$.

$\int\frac{dz}{y}$ $\int\frac{1}{z}dz$ $\frac{4}{2\pi i}$

$2\pi i\cdot X(\ldots)$
0